

packet in their presence after they check it for explosives and drugs? The T-Cognition system is a terahertz scanner developed by researchers at the Fraunhofer Institute for Physical Measurement Techniques IPM in Kaiserslautern in collaboration with Hübner GmbH & Co. KG in Kassel, which helps scan the content of packages or envelopes without having to open them. In specific, the system helps recognise parcels containing hazardous, flammable materials such as explosives and illicit drugs.

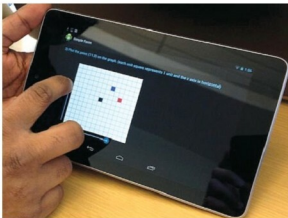
The scientists chose to use waves in the terahertz range for scanning, because this lies midway between microwave and infrared in the electromagnetic spectrum, and thus combines the advantages of both. These low-energy frequency waves can easily penetrate paper, wood, lightweight fabrics, plastics and ceramics. The characteristic spectra generated by them can be analysed using intelligent software to accurately detect the type of material they travel through. Also, since terahertz waves are non-ionising, they are safe to use without any special shields.

Although very useful for mail scanning, this nascent technology can also be used for other security applications such as industrial security, airport security, and so on.

New light for scouting fingerprints

When we say tech in security, it is not just about high-tech video surveillance and cloud-based analysis. Sometimes, simple breakthroughs can be big, such as this new UV light developed at the University of Leicester to identify fingerprints on thermal paper used for printing receipts, ATM bills, etc—a key requirement for solving several financial crimes.

Historically, it has been a challenge to visualise fingerprints on thermal paper, as the solvent used in the chemical process turns the whole paper black. A technology called Hot Print System (HPS) developed some years ago helped read fingerprints on thermal paper by applying heat to the paper; however this too turned out to be



LatentGesture profiles each user and stores their 'touch signature' (Courtesy: Georgia Institute of Technology)

problematic as the quality of thermal paper varied from country to country.

This new technology, on the other hand, uses a specifically-tailored UV light to illuminate the thermal paper and create faint images of fingerprints. This enables quick, non-invasive and speculative examination of fingerprints. The fine images produced by the UV light can then be enhanced and converted to digital form using the earlier HPS system.

Knowing you intrinsically

People you are close to do not need an ID to recognise you! They know you by who you are, what you do, and why you do things. Considering that the mobile phone has now become many a man's closest confidante, have you ever wished your phone too could recognise you instinctively, without needing a password? Perhaps they could,

in the future, thanks to a new security system developed by researchers at the Georgia Institute of Technology.

The new system continuously monitors how a user taps and swipes a mobile device, and tries to understand the patterns in these movements. If a

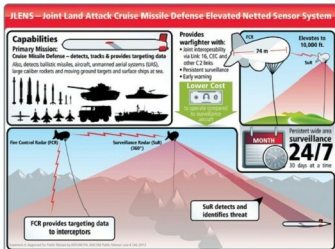
stranger uses the device, the system recognises that the movements do not match the owner's normal tendencies, and can be programmed to take further action such as locking the device or alerting somebody.

Called LatentGesture, the system was tested at the lab using 20 users. As the users do various tasks like tap buttons, swipe sliders or check boxes, LatentGesture de-

velops a profile of each person, what they call a person's 'touch signature.' For every device, the system can store five touch signatures—the owner's and four authorised users'. Whenever somebody uses the mobile device, it automatically identifies whether it is the owner or one of the other authorised users, and allows all operations they are authorised to do. If a stranger uses the system, an alert is issued or the device is locked. The system, tested on Android devices, was nearly 98 percent accurate on a smartphone and 97 percent correct on tablets.

Funny looking blimps, serious about security

Early this year, the US Government announced that it would be deploying two giant security blimps over Maryland around October. The blimps would carry



A bird's eye view of the JLENS (Courtesy: Raytheon)